

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 3a: Introduction: Random variables

Prof. Oke

UMassAmherst

College of Engineering

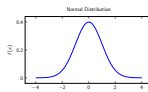
September 29, 2026

Overview of Module 3

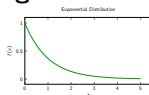
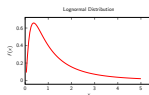
Overview

- Lecture 3a: Introduction: Random Variables

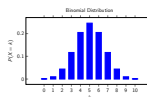
- Lecture 3b: **Normal Distribution**



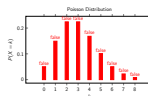
- Lecture 3c: **Lognormal and Exponential Distributions**



- Lecture 3d: **Binomial Distribution**



- Lecture 3e: **Poisson Distribution**



- Lecture 3f: Joint Distributions and further topics

Objectives and outline of today's lecture

- Understand random variables
- Distinguish between discrete and continuous random variables
- Compute measures of centrality and dispersion, as well as sketch PMFs, PDFs and CDFs

① Introduction to random variables

② Probability distribution of r.v.

③ Discrete r.v.'s

④ Continuous r.v.'s

⑤ Outlook

⑥ Appendix

Random variables

Definitions

- A random variable (r.v.) represents the values of the outcomes in a sample space (e.g. the outcome of die roll: $X = \{1, 2, 3, 4, 5, 6\}$)
- A random variable is a function that uniquely maps events in a sample space to the set of real numbers.

A random variable X may be:

- *Discrete*
- *Continuous*

Describing random variables

Central values

- Mean
- Median
- Mode

Measures of dispersion

- Variance
- Standard deviation
- Coefficient of variation (COV)

Probability distribution

A probability distribution governs the values of a random variable.
It can be described by the following functions:

- probability mass function, PMF discrete random variable
- probability density function, PDF continuous random variable
- cumulative distribution function, CDF discrete/continuous random variable

Probability mass function (PMF)

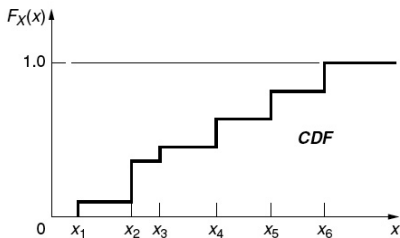
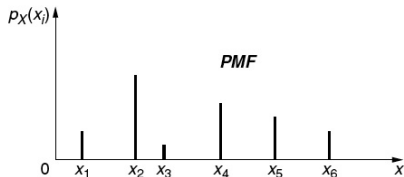
The PMF is given by

$$p_X(x_i) \equiv P(X = x_i) \quad \forall x \quad (1)$$

Note that the symbol $\forall$ means “for all”

Cumulative distribution function (CDF) of discrete random variable

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \sum_{x_i \leq x} P(X = x_i) \\ &= \sum_{x_i \leq x} p_X(x_i) \end{aligned}$$



The probability masses in a PMF sum up to 1.

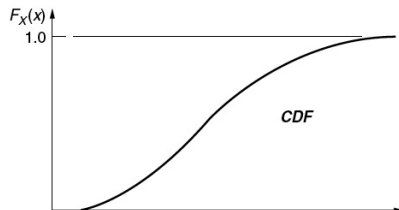
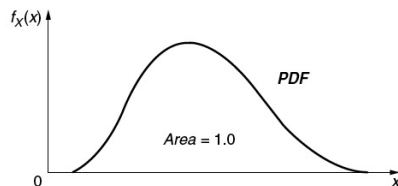
Probability density function (PDF)

The PDF is denoted $f_X(x)$ such that the probability of X in the interval $(a, b]$ is:

$$P(a < X \leq b) = \int_a^b f_X(x) dx \quad (2)$$

Cumulative distribution function (CDF) of continuous random variable

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \int_{-\infty}^x f_X(\tau) d\tau \end{aligned}$$



The total area under a PDF is 1.

Cumulative distribution function (CDF)

The CDF (F_X) of a random variable X is given by

$$F_X \equiv P(X \leq x) \quad \text{for all } x \quad (3)$$

The CDF satisfies the basic axioms of probability:

- ① $F_X(-\infty) = 0$ and $F_X(\infty) = 1$
- ② $F_X(x) \geq 0 \quad \forall x$ and is nondecreasing with x .
- ③ $F_X(x)$ is continuous to the right with x .

Relationship between CDF and PDF

The PDF is the derivative of the CDF:

$$f_X(x) = \frac{dF_X(x)}{dx} \quad (4)$$

while the CDF is the integral of the PDF:

$$F_X(x) = \int_{-\infty}^x f_X(\tau) d\tau \quad (5)$$

Example 1: Operating condition of bulldozers

Each of 3 bulldozers equally likely to operational or nonoperational after 6 months. Plot the PMF and CDF of the random variable X which represents the operating condition of the bulldozers after 6 months.

- Let the outcomes be O (operational) and N (nonoperational)
- There are $2 \times 2 \times 2 = 8$ possibilities:

- 1 OOO
- 2 ONN
- 3 NON
- 4 NOO
- 5 ONN
- 6 NON
- 7 NOO
- 8 NNN

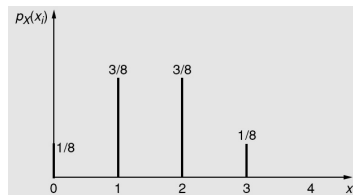


Figure: PMF

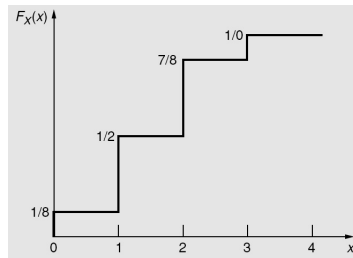


Figure: CDF

Further derivations

① Continuous case:

$$P(a < X \leq b) = \int_{-\infty}^b f_X(x) dx - \int_{-\infty}^a f_X(x) dx \quad (6)$$

② Discrete case:

$$P(a < X \leq b) = \sum_{x_i \leq b} p_X(x_i) - \sum_{x_i \leq a} p_X(x_i) \quad (7)$$

③ For all random variables:

$$P(a < X \leq b) = F_X(b) - F_X(a) \quad (8)$$

Mean and variance

Mean

Weighted average or expected value

$$\mathbb{E}(X) = \sum_i x_i p_X(x_i) \quad \text{discrete case} \quad (9)$$

Variance

In the discrete case:

$$\mathbb{V}(X) = \sum_i (x_i - \mu_X)^2 p_X(x_i) \quad (10)$$

Expanding results in:

$$\mathbb{V}(X) = \mathbb{E}(X^2) - \mu_X^2 \quad (11)$$

Measures of dispersion (cont.)

Standard deviation

The standard deviation is convenient as it has the same unit as the random variable:

$$\sigma_X = \sqrt{\mathbb{V}(X)} \quad (12)$$

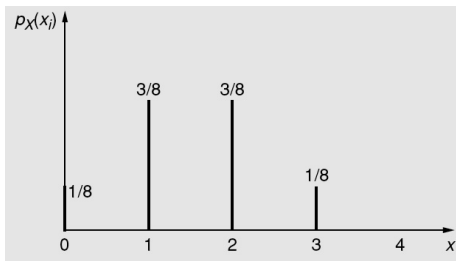
Coefficient of variation

The COV gives the deviation relative to the mean. It is unitless.

$$\delta_X = \frac{\sigma_X}{\mu_X} \quad (13)$$

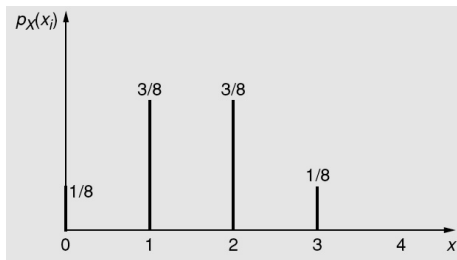
Example 2: Bulldozers revisited

You are given the PMF (probability mass function) of the operating condition X of bulldozers after 6 months.



Find the mean, variance, standard deviation and coefficient of variation of X .

Example 2: Bulldozers revisited (cont.)



- (a) Mean: $\mu_X = \mathbb{E}(X) = 0 \left(\frac{1}{8}\right) + 1 \left(\frac{3}{8}\right) + 2 \left(\frac{3}{8}\right) + 3 \left(\frac{1}{8}\right) = 1.5$.
- (b) Variance: $\mathbb{V}(X) = [0^2 \left(\frac{1}{8}\right) + 1^2 \left(\frac{3}{8}\right) + 2^2 \left(\frac{3}{8}\right) + 3^2 \left(\frac{1}{8}\right)] - (1.5)^2 = 0.75$
- (c) Standard deviation: $\sigma_X = \sqrt{0.75} = 0.866$
- (d) Coefficient of variation: $\delta_X = \frac{0.866}{1.50} = 0.577$

Mean and variance

These include the mean, median and mode.

- Mean: weighted average or expected value

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx \quad \text{continuous case} \quad (14)$$

Variance

In the continuous case:

$$\mathbb{V}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx \quad (15)$$

Expanding both equations results in:

$$\mathbb{V}(X) = \mathbb{E}(X^2) - \mu_X^2 \quad (16)$$

Example 3: Loaded beam

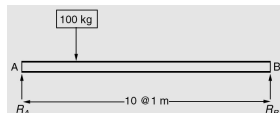


Figure E2.5a Beam AB.

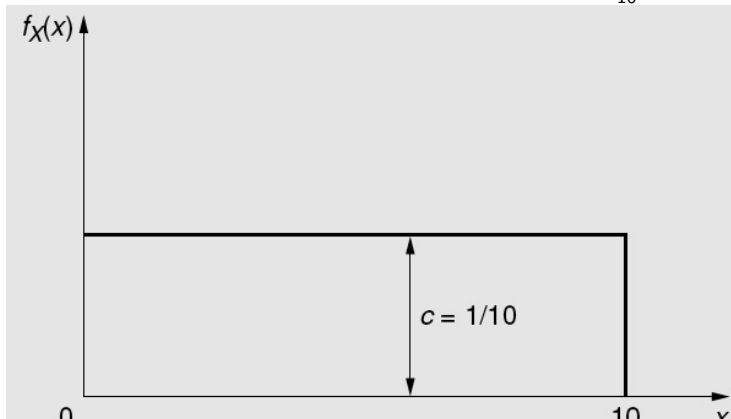
Consider the beam under a 100-kg load. If the load is equally likely to be placed anywhere along the 10m span of the beam, then the PDF of the load position X is **uniformly distributed** in $0 < x \leq 10$, i.e.:

$$f_X(c) = \begin{cases} c & 0 < x \leq 10 \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

- Plot the PDF of X .
- Solve the integral for the CDF and plot.
- Find $P(2 < X \leq 5)$.

Example 3: Loaded beam (cont.)

- (a) The area under the PDF must be 1. Thus, c must be $\frac{1}{10}$.



Example 3: Loaded beam (cont.)

(b) The CDF is given by:

$$F_X = \int_0^x c dx = cx = \frac{x}{10} \quad 0 < x \leq 10$$

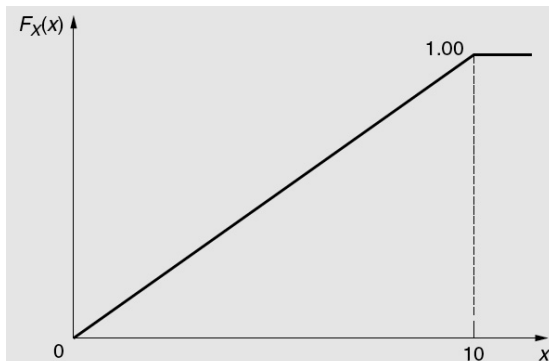


Figure E3.2b CDF of X.

Example 3: Loaded beam (cont.)

(c) To compute $P(2 < X \leq 5)$, we use the CDF:

$$\begin{aligned} P(2 < X \leq 5) &= F_X(5) - F_X(2) \\ &= \frac{5 - 2}{10} = 0.3 \end{aligned}$$

Recap

- Random variables
- Probability mass/density function (PMF/PDF) and cumulative distribution function (CDF)
- Measures of centrality
- Measures of dispersion

Reading

- Open Intro Statistics Section 3.4 (Random variables)
- Open Intro Statistics Section 3.5 (Continuous distributions)

Skewness

The skewness or symmetry of a distribution is measured by the third central moment:

In the discrete case:

$$\mathbb{E}(X - \mu_X)^3 = \sum_i (x_i - \mu_X)^3 p_X(x_i) \quad (18)$$

In the continuous case:

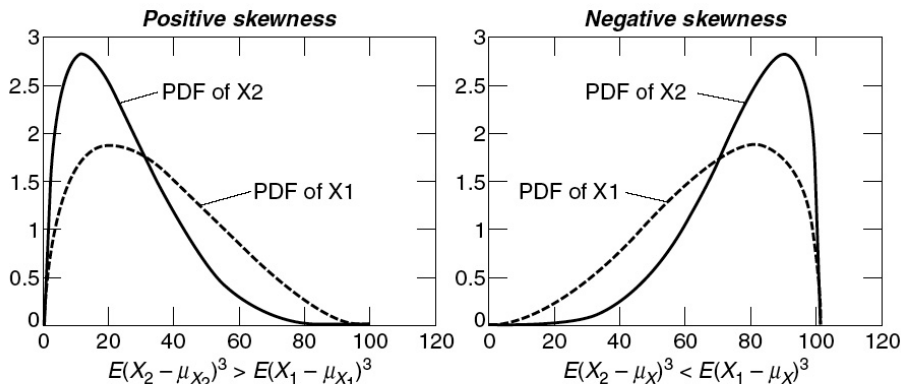
$$\mathbb{E}(X - \mu_X)^3 = \int_{-\infty}^{\infty} (x - \mu_X)^3 f_X(x) dx \quad (19)$$

For convenience, the skewness coefficient is also used (unitless):

$$\theta = \frac{\mathbb{E}(X - \mu_X)^3}{\sigma^3} \quad (20)$$

Skewness (cont.)

- Positive skewness is characterized by a long right tail (right-skewed)
- Negative skewness is characterized by a long left tail (left-skewed)



Kurtosis

This is the measure of peakedness in a distribution. It is the fourth central moment:

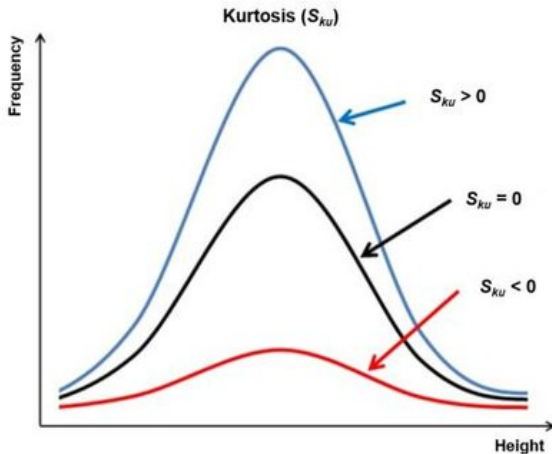
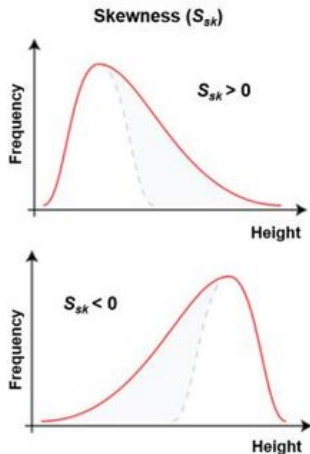
In the discrete case:

$$\mathbb{E}(X - \mu_X)^4 = \sum_i (x_i - \mu_X)^4 p_X(x_i) \quad (21)$$

In the continuous case:

$$\mathbb{E}(X - \mu_X)^4 = \int_{-\infty}^{\infty} (x - \mu_X)^4 f_X(x) dx \quad (22)$$

Skewness vs. kurtosis



Source: Bonyar, A (2015) "Application of localization factor for the detection of tin oxidation with AFM" DOI: 10.1109/SIITME.2015.7342289

Generalized expectation

The mathematical expectation can be defined for a function g of random variable X :

$$E[g(X)] = \sum_i g(x_i) p_X(x_i) \quad \text{discrete case} \quad (23)$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad \text{continuous case} \quad (24)$$